



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO ENERGY POINT

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

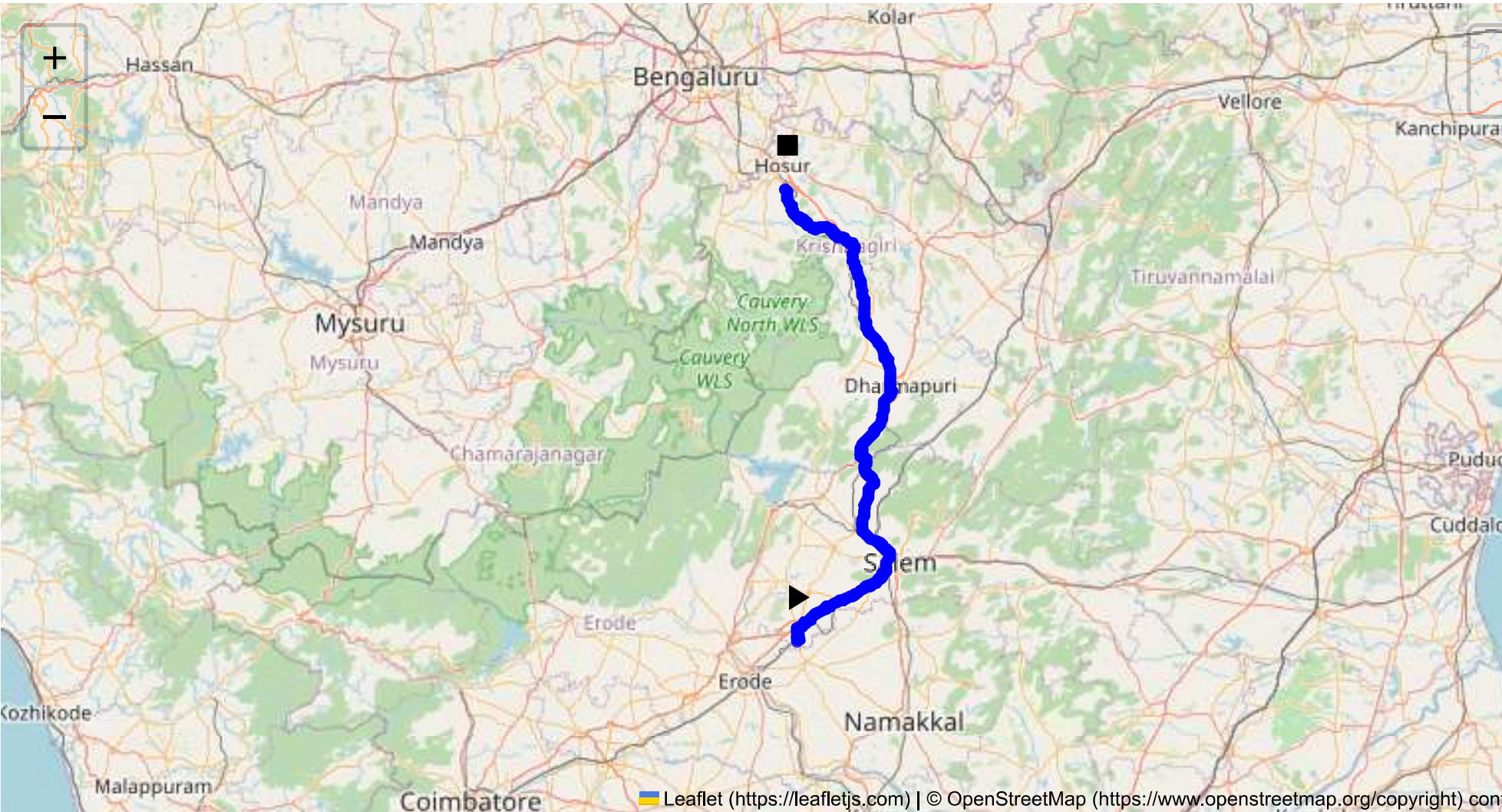
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 180.97 km
Estimated Duration: 3.5 hours
Adjusted Duration (Heavy Vehicle): 4.4 hours
Start: (11.4381, 77.8734)
End: (12.66818, 77.839287)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Hosur spans approximately 180.97 kilometers, typically taking about 3.49 hours for heavy vehicles carrying hazardous materials. Key highways involved include NH44 and NH544, connecting rural and industrial zones through parts of Tamil Nadu.

Weather Conditions and Hazards

Tamil Nadu generally experiences tropical weather, with the route subject to:

- **Monsoon season (June to September):** Heavy rainfall causing potential flooding, slick roads, and reduced visibility.
- **Summer months (March to May):** High temperatures may lead to tire blowouts in under-inflated tires.
- **Fog:** Particularly in winter months, potentially affecting visibility in early mornings and late evenings.

Traffic Patterns

- **Peak Hours:** Morning (7:30-9:30 AM) and evening (6:00-8:00 PM) are most congested, particularly in urban fringes and near industrial areas like Hosur.
- **Congestion Zones:** Urban centers and toll plazas along major highways, and entrances to industrial zones.

Road Quality and Infrastructure

- **Highways:** Generally well-maintained but with occasional potholes and uneven pavements, especially after monsoon rains.
- **Rural Roads:** Potential narrower paths with lesser maintenance, higher risk for large vehicles.
- **Infrastructure:** Adequate signage and lighting on major roads, but less on rural detours.

Alternative Routes

- **In Case of Emergencies:** Alternate route via SH20 can bypass major congestion areas but increases travel time. Local roads may also provide detours but require careful navigation to avoid low-quality paths.

Local Regulations on Hazardous Material Transport

- Transport of hazardous materials must comply with local safety standards, including proper labeling, use of certified containers, and permits.
- Restricted movement in certain dense residential and commercial areas during peak hours.

Historical Incidents

- **Reported Incidents:** Historically, accidents involving heavy vehicles typically occur due to brake failure or human error. No significant incidents involving hazardous material spills have been documented recently.

Environmental Considerations

- **Sensitive Areas:** Proximity to agricultural lands requires careful driving to prevent contamination.
- **Wildlife Crossings:** Some rural roads may intersect with areas frequented by local fauna.

Communication Coverage

- **Dead Zones:** Due to rural areas, there might be intermittent mobile network failures. It's advisable to use GPS systems with offline maps.

Emergency Response Times

- **Urban Areas:** Fast response within 30 minutes due to proximity to facilities.
- **Rural Segments:** Could vary, expected up to 1.5 hours due to terrain and road conditions.

Overall Risk Assessment

This route presents moderate risk due to weather, traffic, and road quality factors. Drivers are advised to remain vigilant during peak weather conditions, adhere to traffic regulations, and keep emergency contacts at ready, especially when passing through rural and less accessible areas. Implement regular vehicle maintenance checks and ensure full compliance with hazardous material transport regulations for enhanced safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.46400, 77.87323	30 KM/Hr
5	Turn	Medium	11.46765, 77.87037	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	Medium	11.47986, 77.88397	30 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.53501, 77.95707	30 KM/Hr
11	Turn	Medium	11.53601, 77.95856	30 KM/Hr
12	Turn	Medium	11.62403, 78.10893	30 KM/Hr
13	Turn	Medium	11.62917, 78.11648	30 KM/Hr
14	Turn	High	11.64180, 78.11808	15 KM/Hr
15	Turn	High	11.64325, 78.11993	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
16	Turn	Medium	11.67865, 78.12803	30 KM/Hr
17	Turn	Medium	11.68201, 78.12639	30 KM/Hr
18	Turn	Medium	11.69519, 78.10775	30 KM/Hr
19	Turn	Medium	11.73958, 78.05105	30 KM/Hr
20	Turn	Medium	11.75835, 78.05016	30 KM/Hr
21	Turn	Medium	11.77934, 78.05765	30 KM/Hr
22	Turn	Medium	11.86847, 78.08367	30 KM/Hr
23	Turn	Medium	11.87942, 78.08197	30 KM/Hr
24	Turn	Medium	11.91177, 78.05854	30 KM/Hr
25	Turn	Medium	11.92585, 78.06507	30 KM/Hr
26	Turn	Medium	11.93319, 78.06419	30 KM/Hr
27	Turn	Medium	11.93557, 78.06103	30 KM/Hr
28	Turn	Medium	11.93704, 78.05402	30 KM/Hr
29	Turn	Medium	11.95664, 78.04781	30 KM/Hr
30	Turn	High	11.95818, 78.04978	15 KM/Hr
31	Turn	Medium	11.96090, 78.04976	30 KM/Hr
32	Turn	High	11.96399, 78.05135	15 KM/Hr
33	Turn	Medium	11.96496, 78.04988	30 KM/Hr
34	Turn	High	11.96675, 78.04939	15 KM/Hr
35	Turn	Medium	11.96791, 78.05070	30 KM/Hr
36	Turn	Medium	12.11983, 78.13276	30 KM/Hr
37	Turn	Medium	12.20409, 78.12245	30 KM/Hr
38	Turn	Medium	12.50505, 78.03133	30 KM/Hr
39	Blind Spot	Blind Spot	12.51384, 78.03205	10 KM/Hr
40	Turn	Medium	12.51372, 78.03018	30 KM/Hr
41	Turn	Medium	12.51637, 78.02568	30 KM/Hr
42	Turn	Medium	12.52254, 78.02328	30 KM/Hr
43	Turn	Medium	12.52712, 78.00951	30 KM/Hr
44	Turn	Medium	12.52900, 78.00856	30 KM/Hr
45	Turn	Medium	12.54948, 77.97277	30 KM/Hr
46	Turn	High	12.55078, 77.97257	15 KM/Hr
47	Turn	Medium	12.55317, 77.96530	30 KM/Hr
48	Turn	Medium	12.56517, 77.95315	30 KM/Hr
49	Turn	Medium	12.56247, 77.92184	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
50	Turn	Medium	12.56413, 77.91144	30 KM/Hr
51	Turn	Medium	12.58134, 77.89471	30 KM/Hr
52	Turn	Medium	12.58412, 77.88892	30 KM/Hr
53	Turn	Medium	12.58362, 77.88778	30 KM/Hr
54	Turn	Medium	12.58605, 77.88057	30 KM/Hr
55	Turn	Medium	12.59032, 77.87684	30 KM/Hr
56	Turn	Medium	12.59273, 77.86889	30 KM/Hr
57	Turn	Medium	12.62182, 77.85190	30 KM/Hr
58	Turn	High	12.65481, 77.84575	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
6	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
8	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
9	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
10	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
11	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
13	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
18	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
19	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
20	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
22	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
23	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
24	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
25	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
26	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
27	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
30	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
31	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
32	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
33	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
34	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
12	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
16	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
17	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
21	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
28	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
29	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
35	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
36	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47986, 77.88397



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62403, 78.10893



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62917, 78.11648



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64180, 78.11808



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64325, 78.11993



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67865, 78.12803



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68201, 78.12639



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69519, 78.10775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73958, 78.05105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75835, 78.05016



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77934, 78.05765



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86847, 78.08367



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91177, 78.05854



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92585, 78.06507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93319, 78.06419



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93557, 78.06103



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93704, 78.05402



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95664, 78.04781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96090, 78.04976



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96496, 78.04988



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.11983, 78.13276



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50505, 78.03133



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.51384, 78.03205



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51372, 78.03018



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51637, 78.02568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52254, 78.02328



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52712, 78.00951



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52900, 78.00856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.54948, 77.97277



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.55078, 77.97257



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.55317, 77.96530



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56517, 77.95315



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56247, 77.92184



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56413, 77.91144



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58134, 77.89471



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58412, 77.88892



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58362, 77.88778



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58605, 77.88057



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59032, 77.87684



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59273, 77.86889



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.62182, 77.85190



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.65481, 77.84575